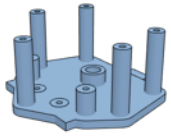
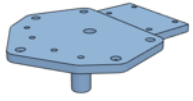
Parts

Print these (STEP in ../STEP). Right-hand build.



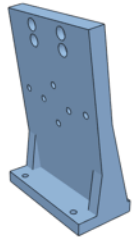
base_bottom

1×



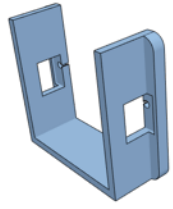
base_top

1×



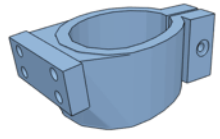
tab_mast

1×



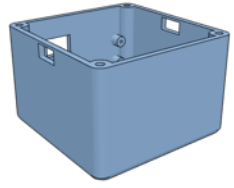
d405_hood

1×



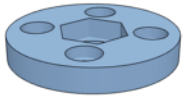
quest_arm_right

1×



ebox_cover

1×



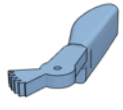
servo_coupler

1×



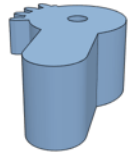
handle_a_rh

1×



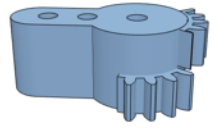
handle_b_rh

1×



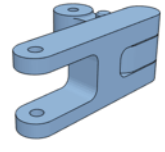
knuckle_gear_left

1×



knuckle_gear_right

1×



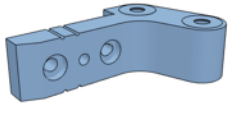
outer_link_2f85

2×



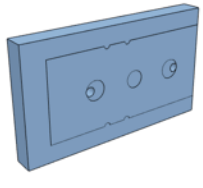
inner_knuckle_2f85

2×



inner_finger_2f85

2×



pad_core_2f85

2×



pad_skin_2f85

2×



grip_tpu_a_rh

1×



grip_tpu_b_rh

1×

0





Press each insert in with a soldering iron at ~220 °C, flush with the surface, before assembling.

Heat-set inserts

Part	Inserts	Where
base_bottom	**9 × M3**	5 standoff tops, 4 underneath for the box
base_top	**4 × M3**	the mast's foot
handle_a_*	**1 × M4 + 1 × M3**	axle bolt (top of the hub), stop screw (underneath)
knuckle_gear_left / _right	**2 × M3 each**	the key screws
quest_arm_*	**4 × M3**	the pad on the mast's back

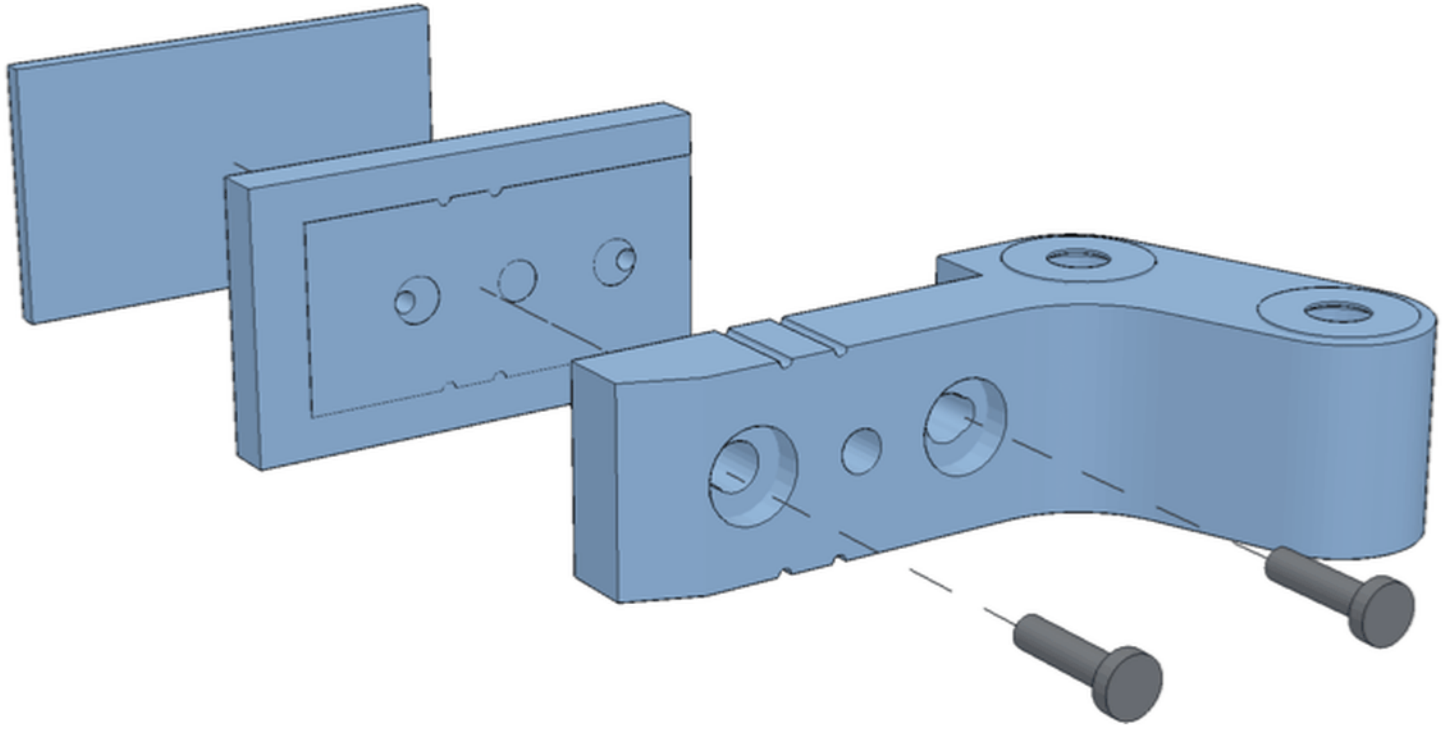
Total: 22 × M3 (M3 × 5.7) and 1 × M4. Glue the TPU grip inserts into their handles (CA glue).

1

Pads

2×

1×	2×
TPU skin: CA glue	M2 × 8 self-tapping

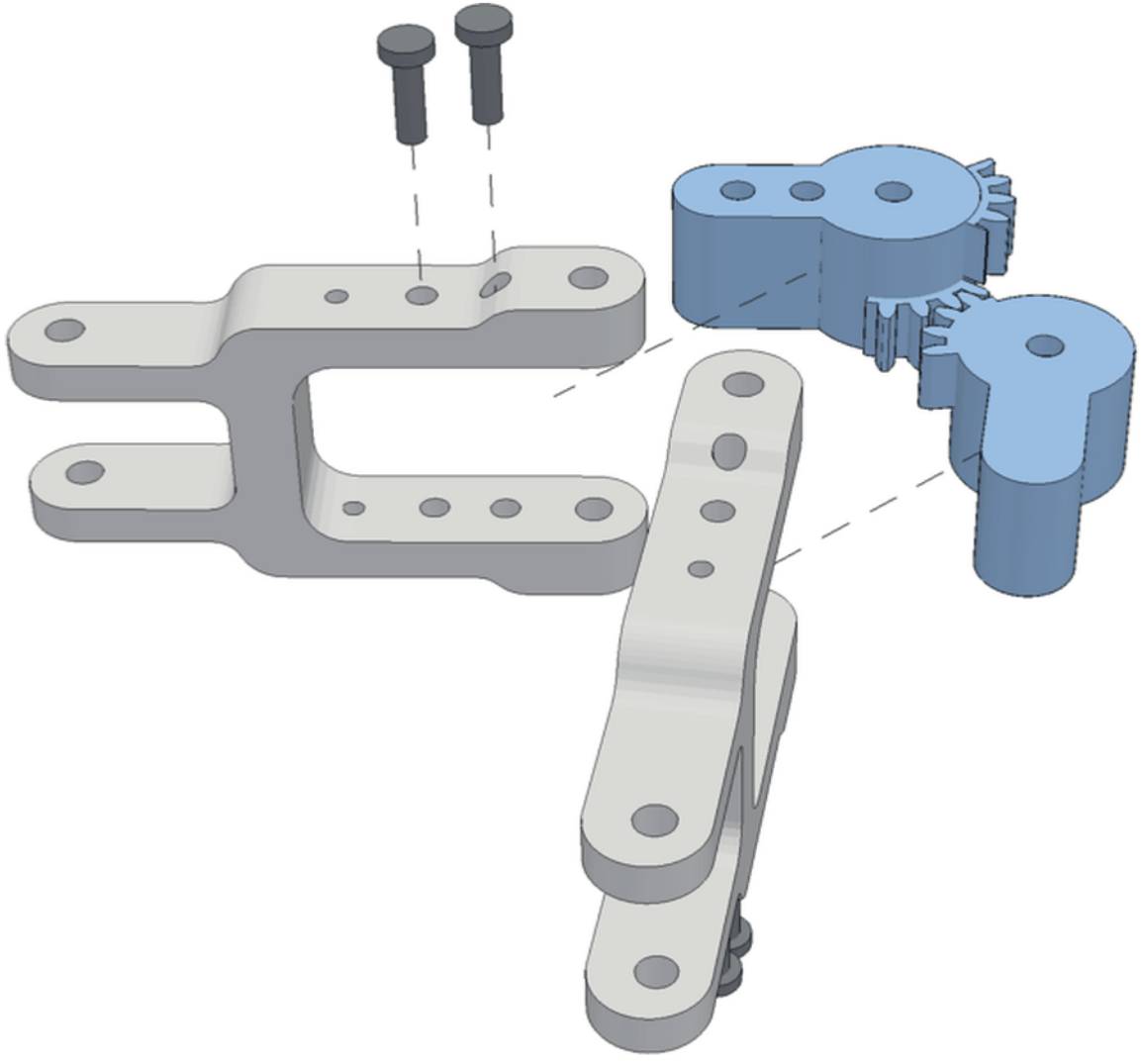


Glue the TPU skin to the core. Screw the core to the inner finger. Make two.

2

Knuckle gears

4×
M3 × 10



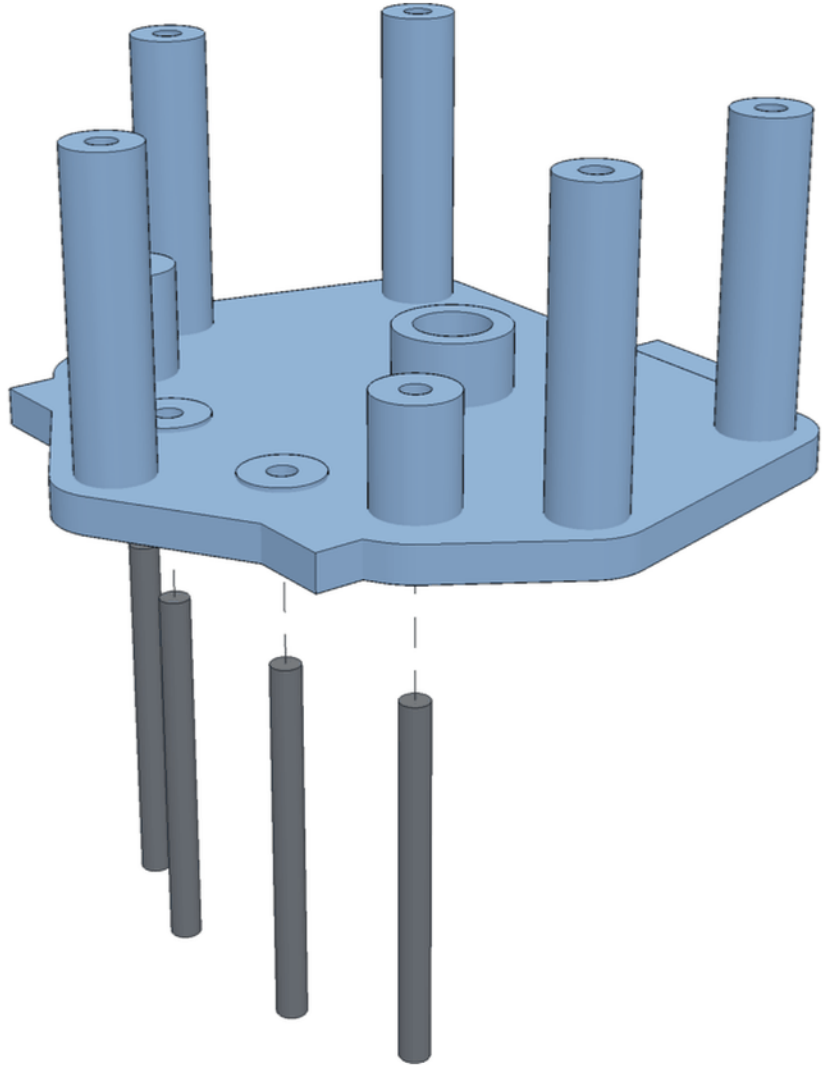
Slide each gear into its inner knuckle's fork. Left gear: screws from below. Right gear: from above.

3

Bottom plate

4× Ø4 × 50 dowel

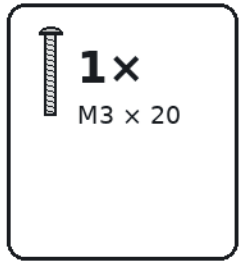
1× 12 × 12 tactile switch

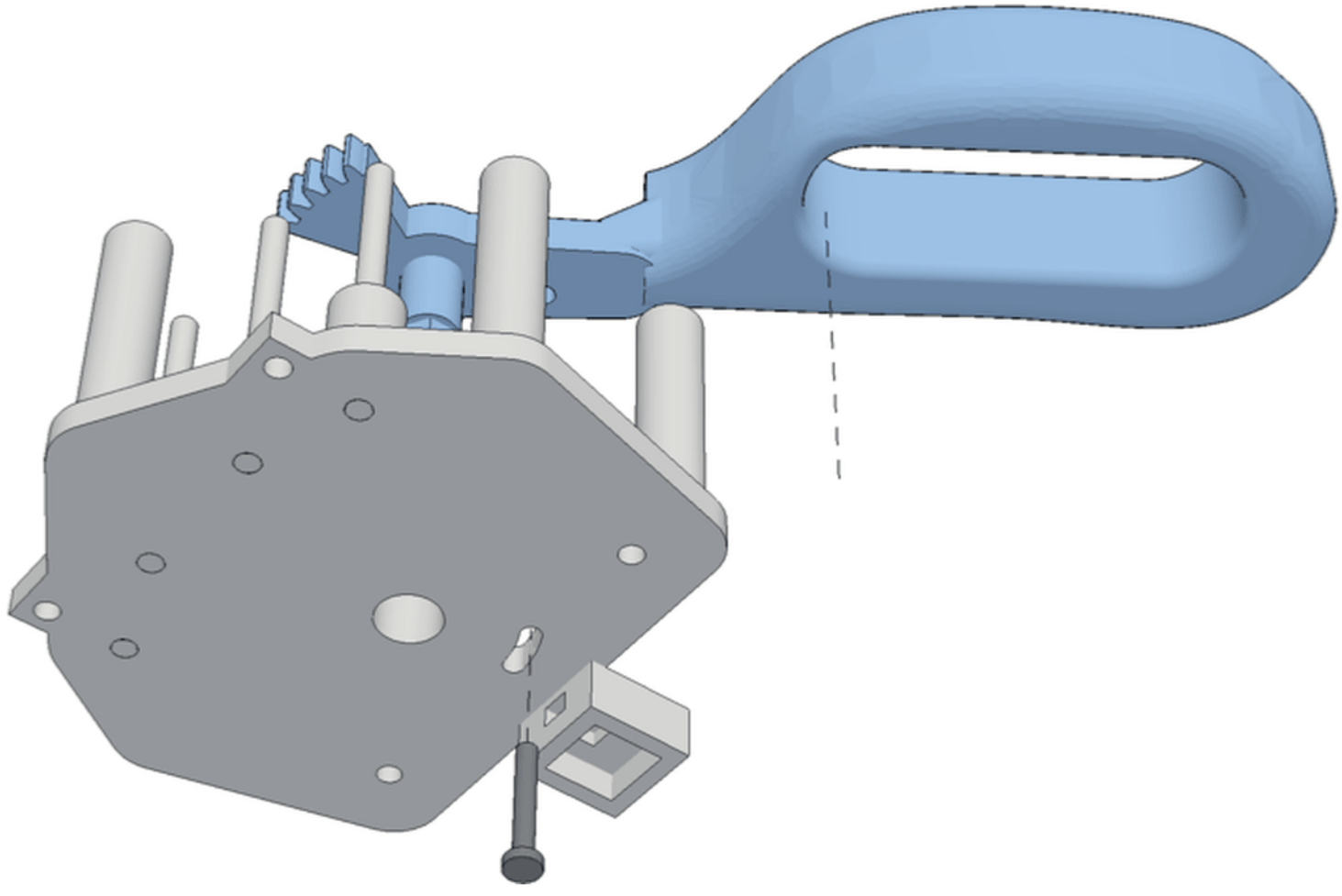


Press the four long dowels into the plate (a vice helps). Push the switch into the pod underneath, button down.

4

Handle A

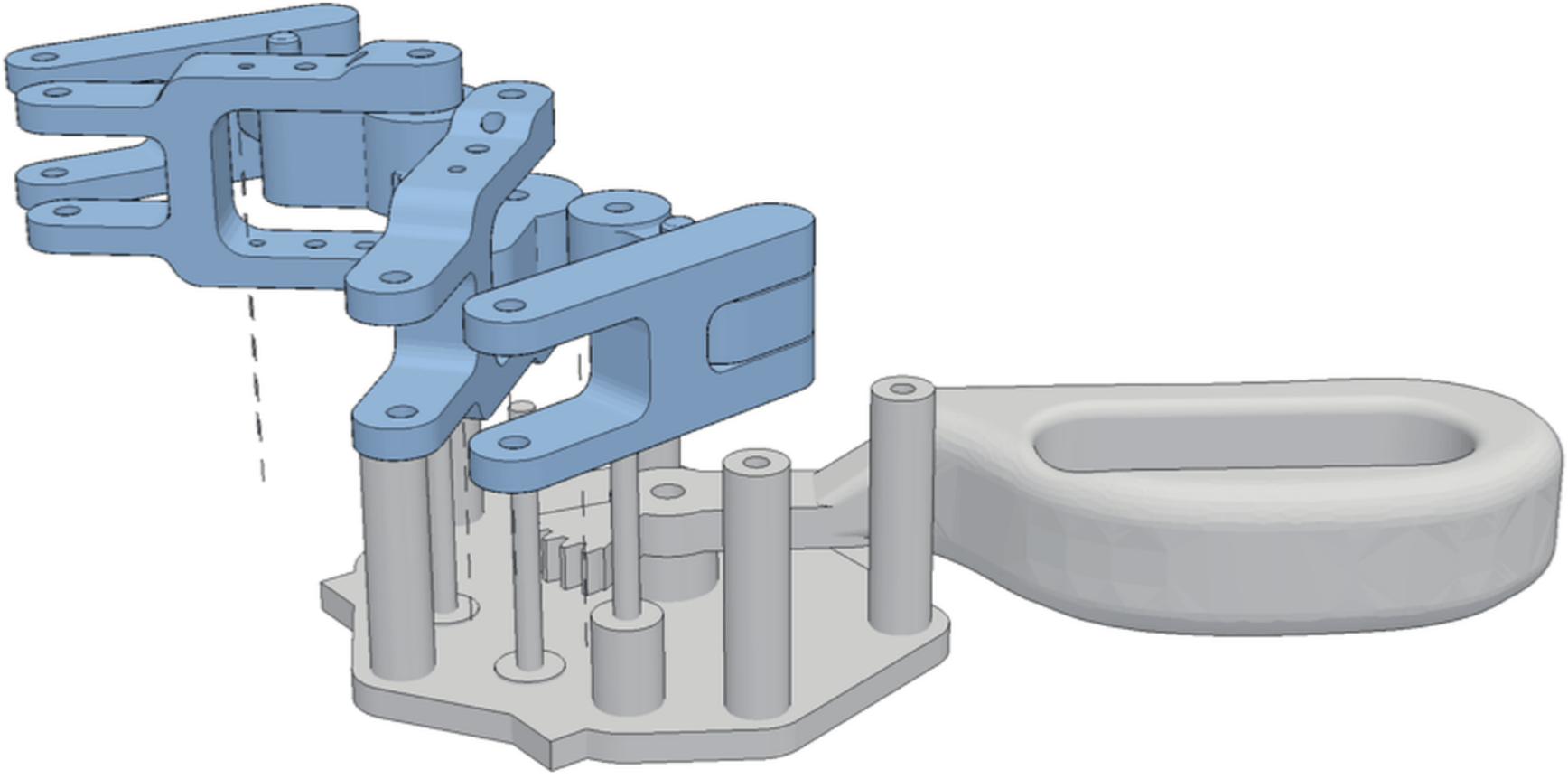




Drop handle A's hub into the bottom plate's bearing. From below, screw the stop screw through the arc slot into handle A.

5

Jaws

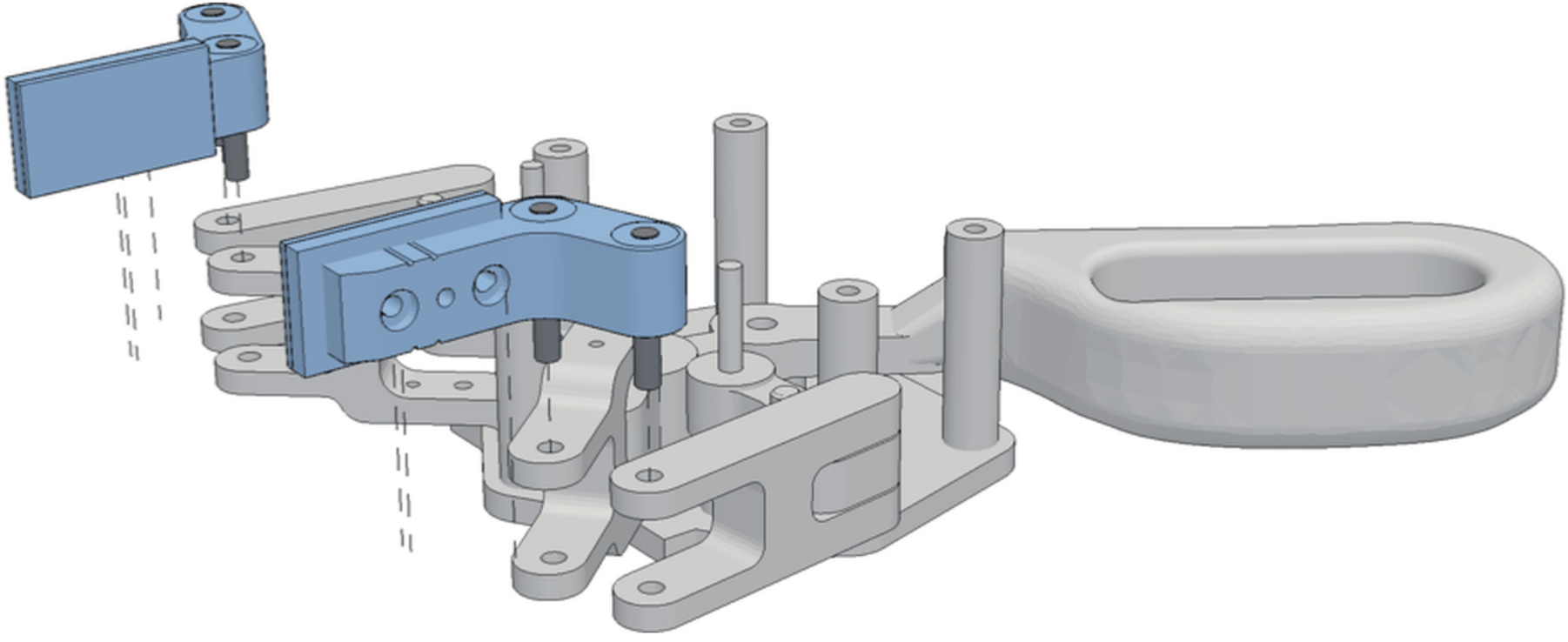


Fully open: hold handle A against its open stop. Slide both outer links and both knuckle + gear assemblies onto the dowels so the gears mesh with each other and with handle A.

6

Fingers

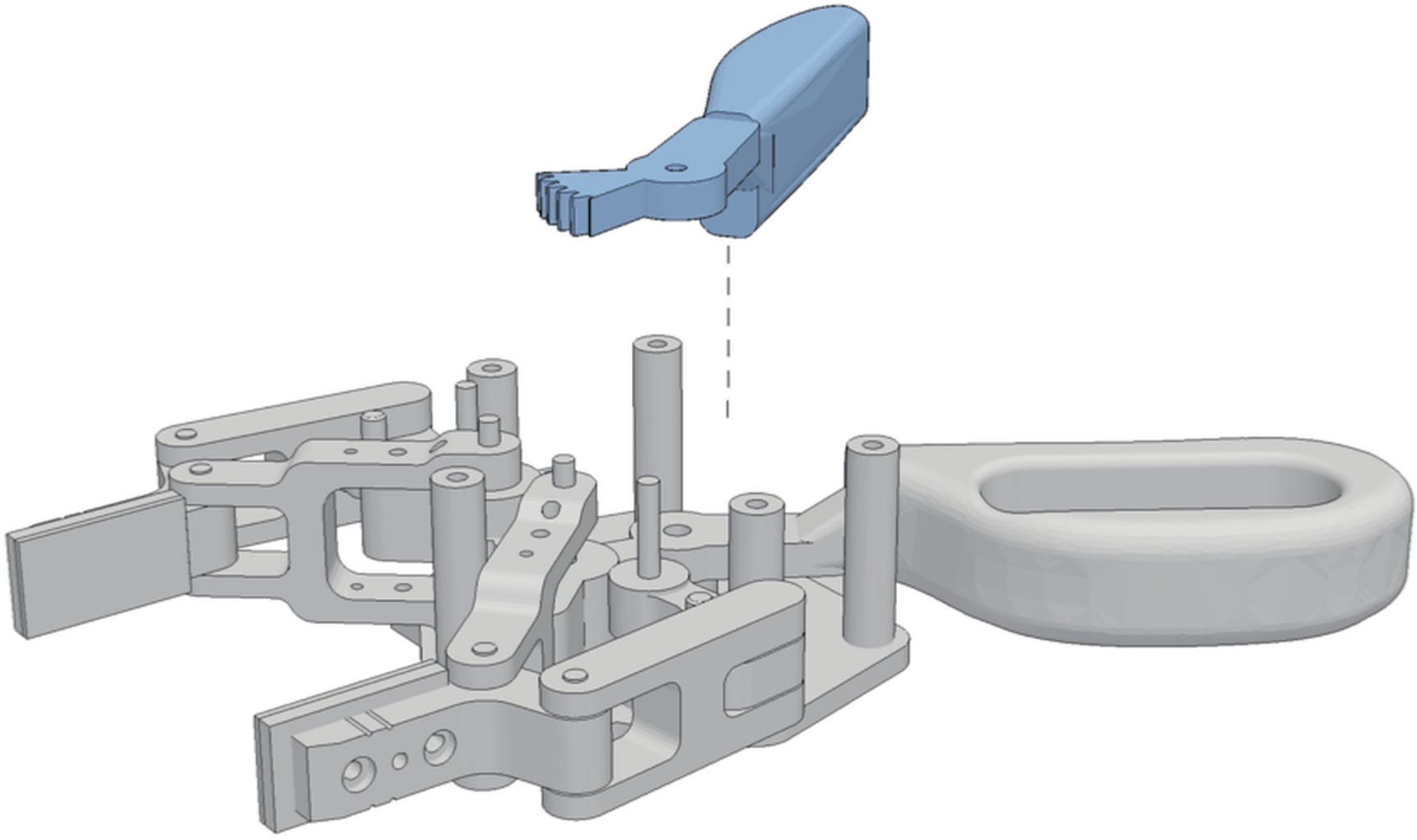
4×
Ø4 × 25 dowel



Fit each finger (with its pad) to its outer link and inner knuckle and pin both joints.

7

Handle B



With the jaws still open, lay handle B on the axle so its teeth mesh with the right gear.

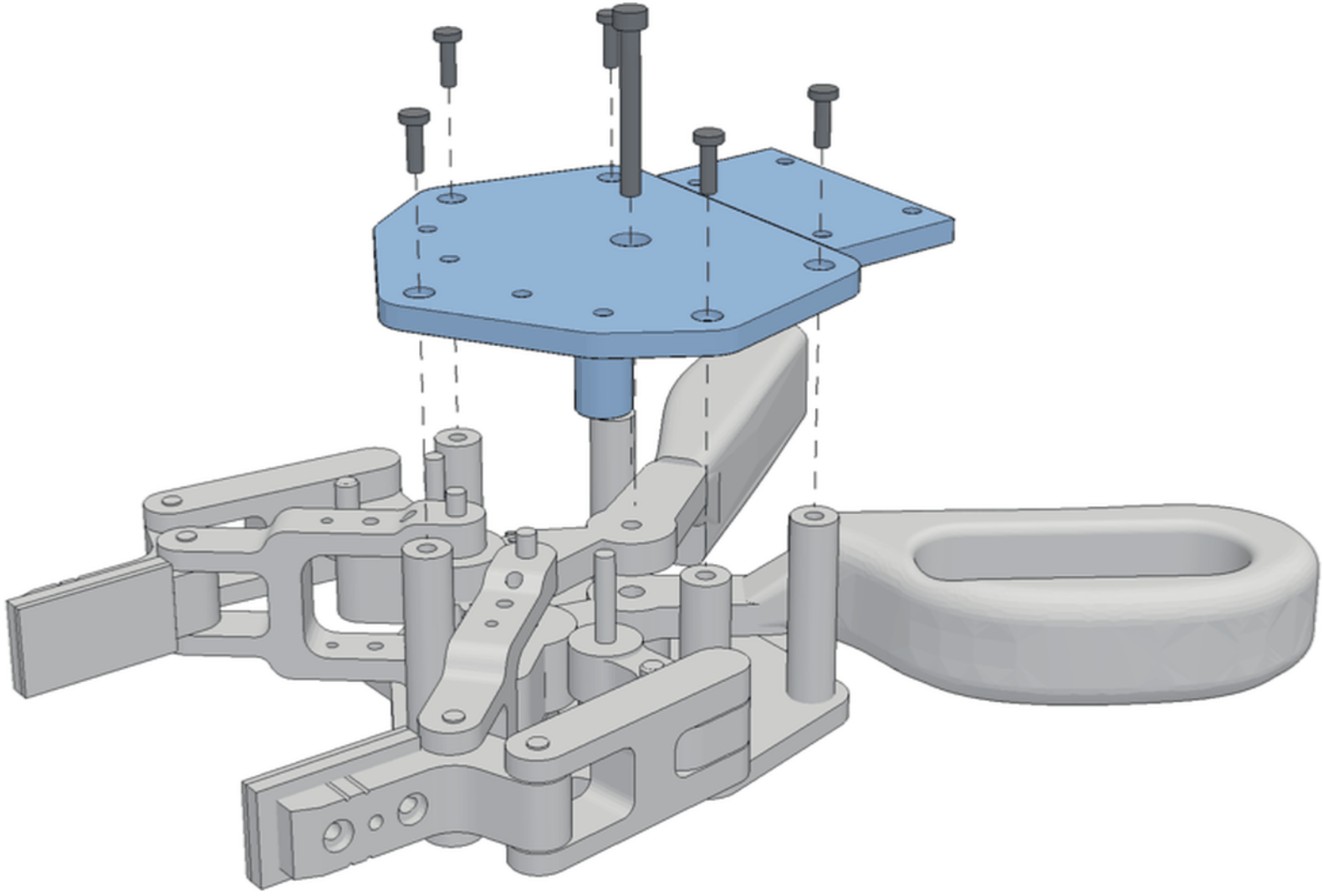
8

Top plate

5× M3 × 10

1× M4 × 35

1× threadlocker on the M4



Fit the top plate over the dowels. Screw it to the five standoffs. Axle bolt through the top plate and handle B into handle A, with threadlocker; snug, so the handles still turn freely.

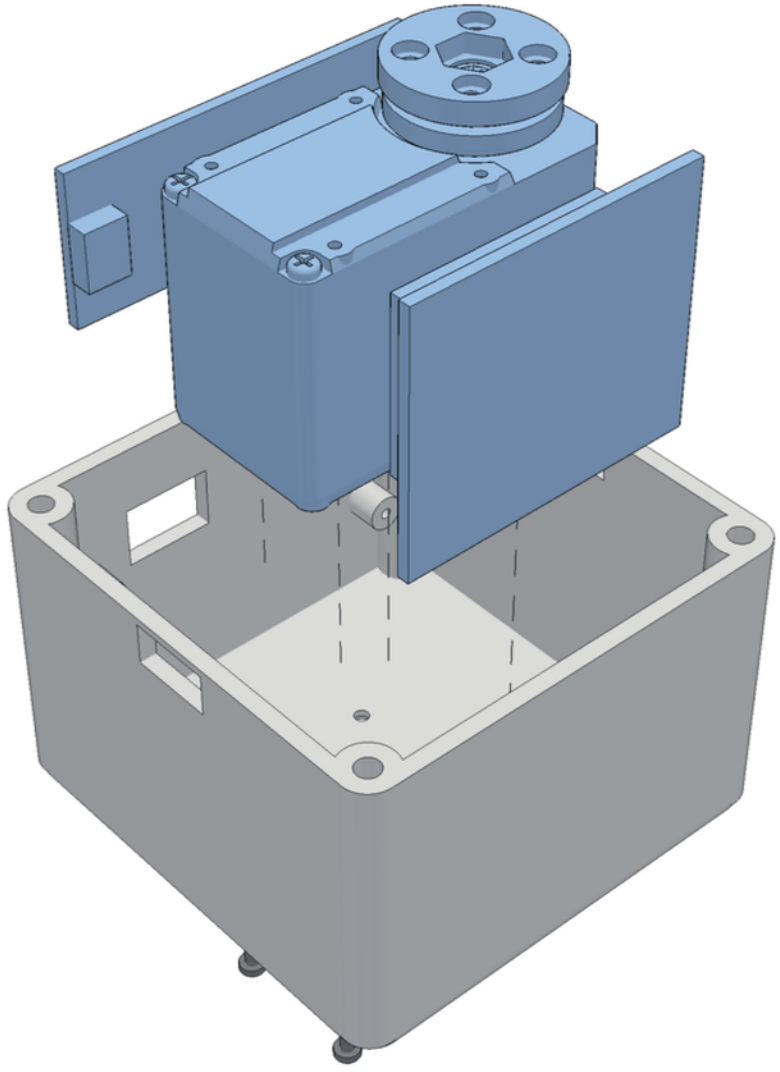
9

Electronics

4× M2 × 8

4× M2 × 6

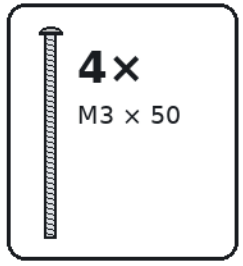
4× M2 × 5 self-tapping

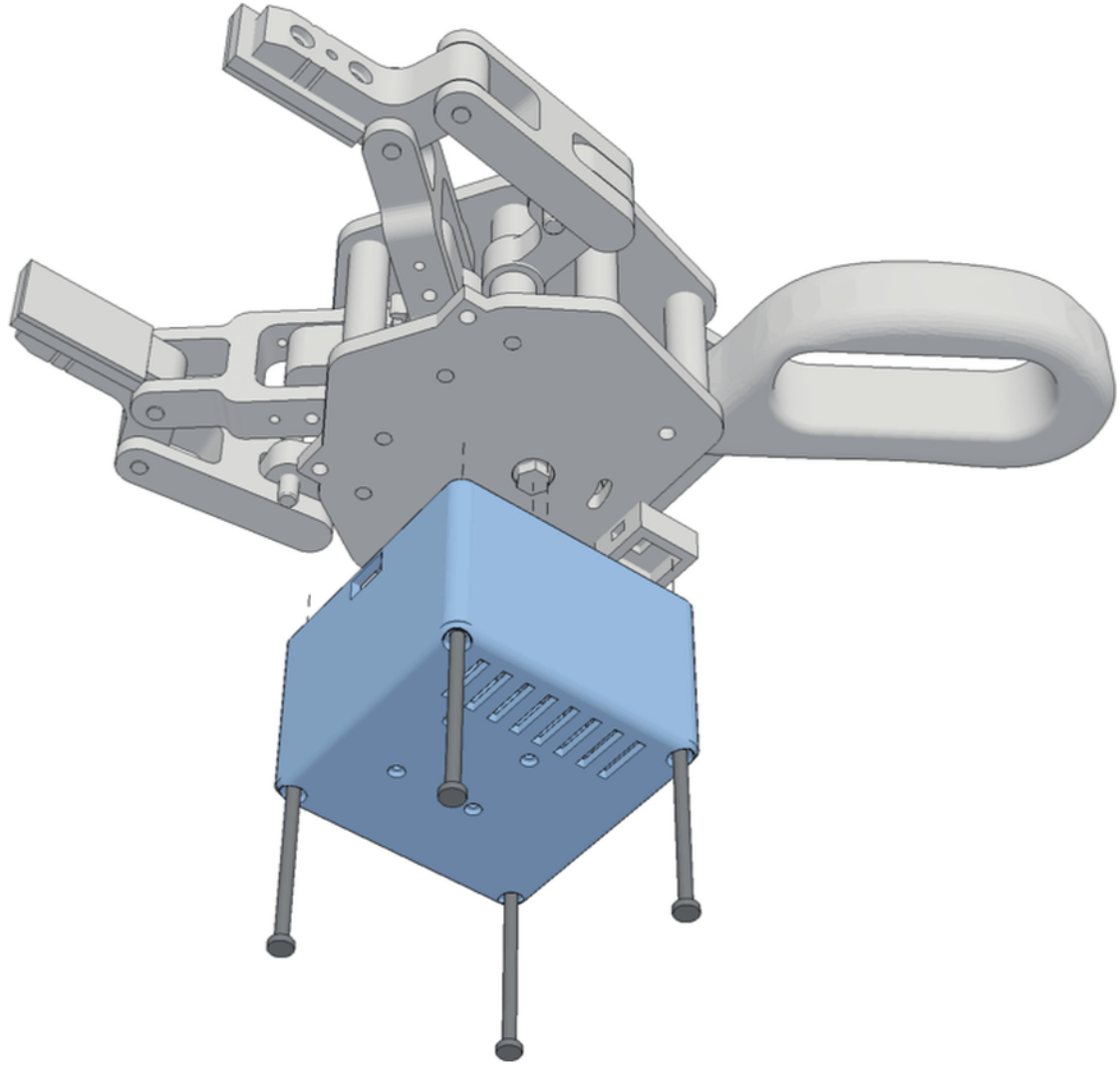


Servo on the box floor (4 × M2 × 8 from below). Coupler on the servo horn (4 × M2 × 6). Pico 2 on its standoffs, hub in its bay, IMU on the back wall. Wire them as in electronics.md.

10

Electronics box



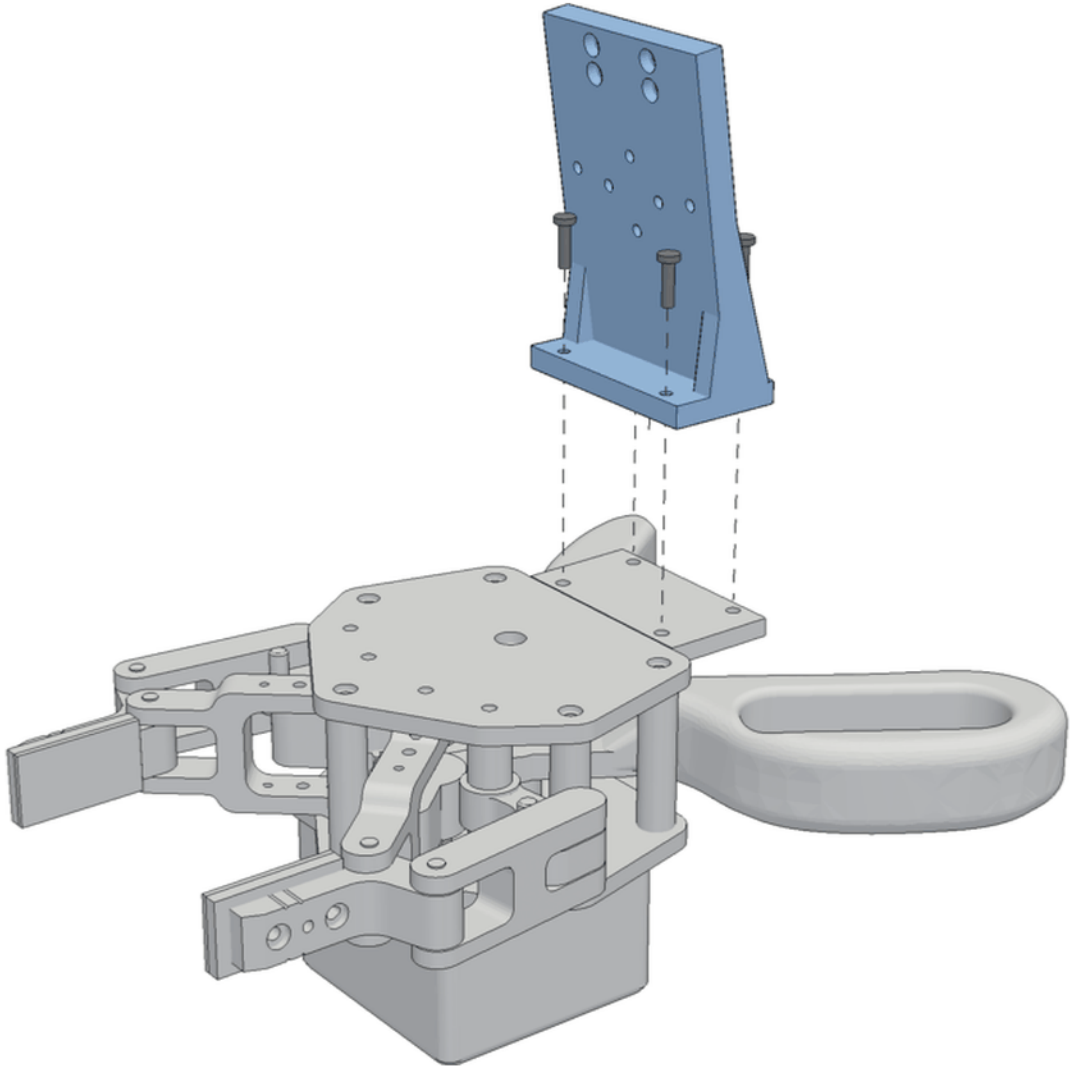


Jaws open, servo at its open position: push the box up so handle A's hex enters the coupler. Screw the box to the bottom plate.

11

Camera mast

4×
M3 × 12

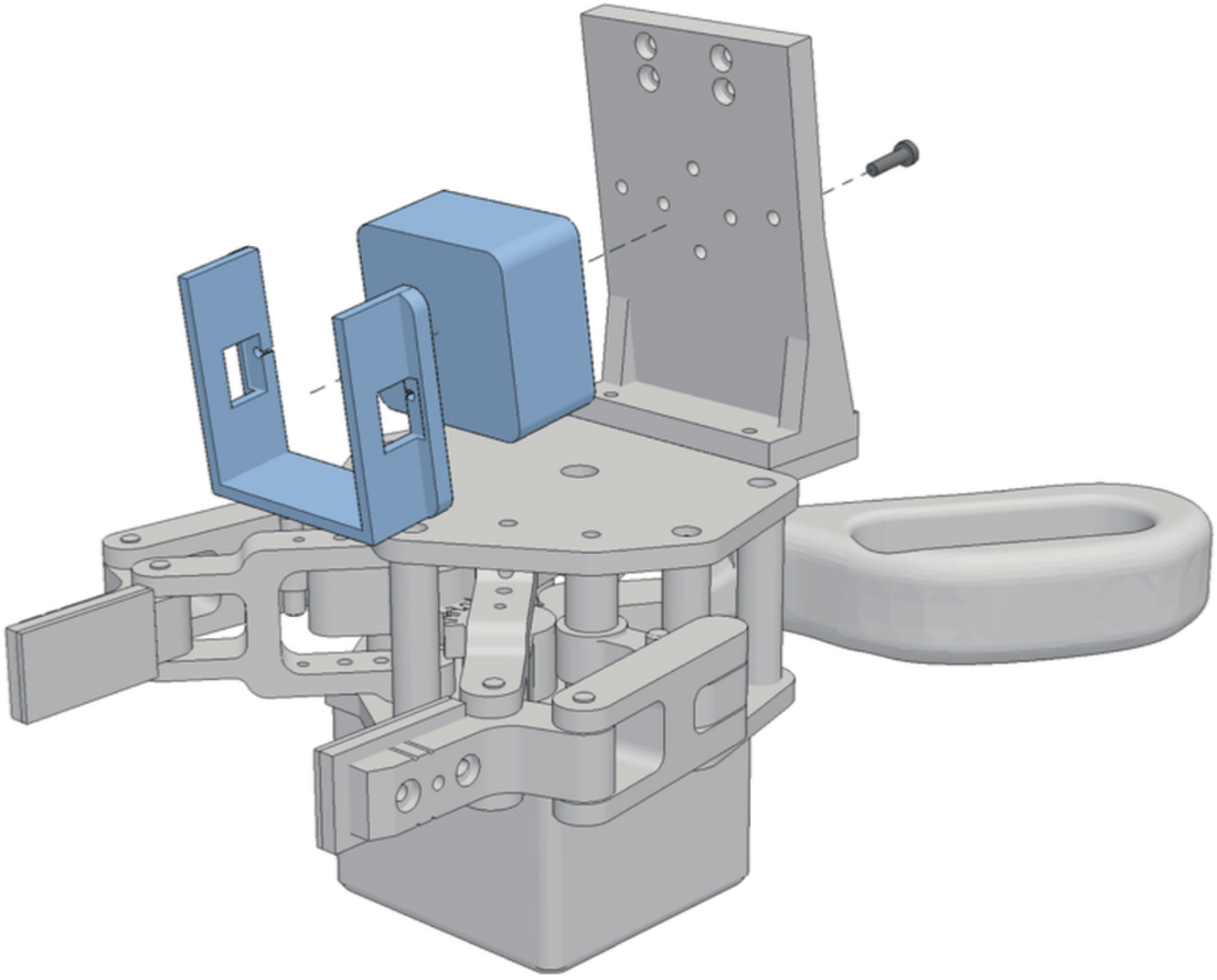


Screw the mast's foot to the top plate.

12

D405

2×
M3 × 12 self-tapping

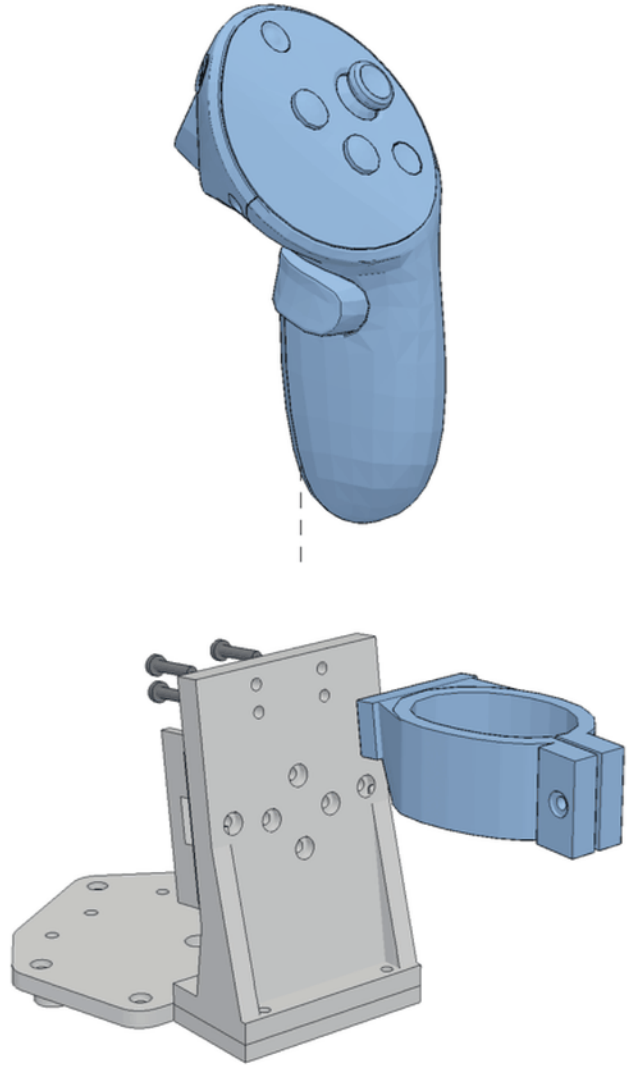


Fix the D405 to the tab exactly as on the robot's camera plate. Put the hood over it and screw it on from behind the mast.

13

Quest controller

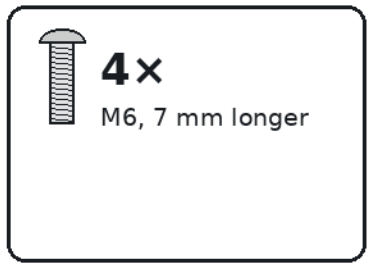
4×	M3 × 16
1×	M3 × 20 + nut



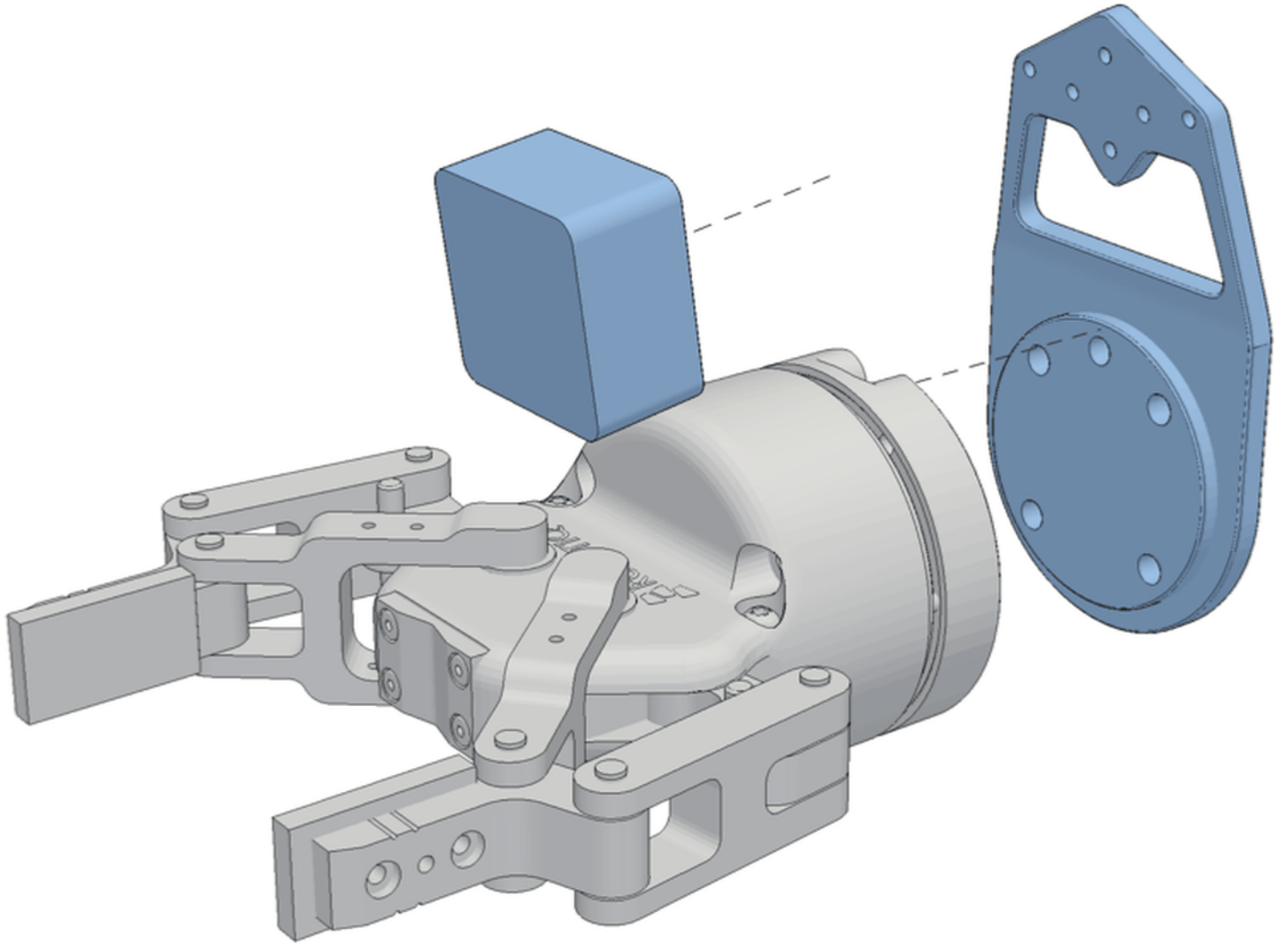
Screw the right Quest arm to the mast's back from the front. Push the right Touch Pro's handle into the collar, trigger forward, and clamp it.

14

Robot



4×
M6, 7 mm longer



On the robot: the Robotiq 2F-85 as it is. Put our camera plate between its coupling and the UR flange (M6 screws 7 mm longer than the coupling's own) and fix the D405 to the plate's tab.